

- 1) An array contains N distinct integers. What is the tightest upper bound time complexity for finding if two elements sum to a target value if the array is unsorted versus if it is pre-sorted?
- **A)** Unsorted: $O(N^2)$, Pre-sorted: $O(N \log N)$
 - **B)** Unsorted: $O(N)$, Pre-sorted: $O(N)$
 - **C)** Unsorted: $O(N \log N)$, Pre-sorted: $O(\log N)$
 - **B)** Unsorted: $O(N)$, Pre-sorted: $O(\log N)$
- 2) Consider a process that inserts N elements into an initially empty Binary Search Tree (BST) where elements arrive in strictly ascending order. What is the time complexity to search for the smallest element in this tree?
- **A)** $O(1)$
 - **B)** $O(\log N)$
 - **C)** $O(N)$
 - **D)** $O(N \log N)$
- 3) Which sorting algorithm maintains $O(N \log N)$ worst-case time complexity while performing operations in-place without requiring extra space proportional to array size?
- **A)** Merge Sort
 - **B)** Quick Sort
 - **C)** Heap Sort
 - **D)** Counting Sort
- 4) A sequence of operations involves pushing elements onto a stack and popping them into a queue. If N elements are processed, what is the overall asymptotic time complexity of emptying the queue?
- **A)** $O(1)$
 - **B)** $O(\log N)$
 - **C)** $O(N)$
 - **D)** $O(N^2)$
- 5) What is the space complexity of a recursive implementation of binary search on a sorted array of size N , assuming stack frame overhead is accounted for?
- **A)** $O(1)$
 - **B)** $O(\log N)$
 - **C)** $O(N)$
 - **D)** $O(N \log N)$

- 6) What is the time complexity of Insertion Sort when applied to an array that is already sorted in reverse order?
- **A)** $O(1)$
 - **B)** $O(N)$
 - **C)** $O(N \log N)$
 - **D)** $O(N^2)$
- 7) When using Quick Sort, what choice of pivot leads to its worst-case time complexity of $O(N^2)$?
- **A)** Always choosing the median element
 - **B)** Always choosing the smallest or largest element
 - **C)** Choosing a random element as the pivot
 - **D)** Choosing the middle element of the current subarray
- 8) Scenario: You are building a feature in a text editor to check whether brackets in a string (e.g., "`{ [ ( ) ] }`") are properly balanced. Which data structure is best suited for this task, and what is the core logic when parsing the string?
- **A)** Queue; push closing brackets to the back and dequeue opening brackets from the front.
 - **B)** Stack; push opening brackets onto the stack and pop to match when encountering a closing bracket.
 - **C)** Array; insert all brackets and perform Binary Search to match pairs.
 - **D)** Hash Table; store each opening bracket with its index as the key.
- 9) Scenario: You are designing a print job management system where documents print in the exact order they are received. Which data structure and operations manage print jobs in order, and how does a Circular Queue solve the structural limitation of a standard array-based queue?
- **A)** Stack with push and pop; a Circular Queue resizes the underlying array dynamically.
 - **B)** Priority Queue with insert and remove; a Circular Queue sorts elements automatically.
 - **C)** Queue with enqueue (back) and dequeue (front); a Circular Queue reuses empty spaces left by dequeued items by wrapping pointers around using modulo arithmetic.
 - **D)** Linked List with add and delete; a Circular Queue eliminates memory leaks.

10) Scenario: You have a small array that is nearly completely sorted (e.g., [1, 2, 4, 3, 5]), with only one adjacent pair out of order. Which algorithm performs best on nearly-sorted data, and why?

- **A)** Selection Sort, because it makes the minimum number of swaps regardless of array order.
- **B)** Insertion Sort, because it achieves nearly $O(N)$ linear performance by shifting elements only when necessary.
- **C)** Bubble Sort, because it swaps non-adjacent elements across the array instantly.
- **D)** Selection Sort, because it scans smaller subarrays at each iteration step.

11) What will be the output of the following code?

```
const user1 = { id: 101, name: "Alice" };  
  
const user2 = { id: 101, name: "Alice" };  
  
console.log(user1 === user2);  
  
console.log(JSON.stringify(user1) === JSON.stringify(user2));
```

12) Predict the console output of this snippet:

```
function scopeTest() {  
  
    console.log(val);  
  
    var val = 100;  
  
    console.log(val);  
  
}  
  
scopeTest();
```

13) console.log(typeof NaN);

```
console.log(NaN === NaN);
```

```
14) for (var i = 0; i < 3; i++) {  
    setTimeout(() => console.log(i), 10);  
}  
  
for (let j = 0; j < 3; j++) {  
    setTimeout(() => console.log(j), 10);  
}
```

```
15) const obj = {  
    data: 42,  
    getData: function() {  
        return this.data;  
    }  
};  
  
const extract = obj.getData;  
  
console.log(extract());
```

```
16) console.log([] == ![]);
```

```
17) console.log("1");  
  
    setTimeout(() => console.log("2"), 0);  
  
    Promise.resolve().then(() => console.log("3"));  
  
    console.log("4");
```

18) `console.log("1");`

`setTimeout(() => console.log("2"), 0);`

`Promise.resolve().then(() => console.log("3"));`

`console.log("4");`

19) `const numbers = [1, 2, 3];`

`const result = numbers.map(num => { num * 2; });`

`console.log(result);`

20) What happens when you execute `Object.create(null)` vs `{}` in terms of prototype chain access (e.g., calling `.toString()`)?

A) Both objects successfully execute `.toString()` and return "[object Object]".

B) `Object.create(null)` returns "[object Object]", while `{}` throws a `TypeError`.

C) `{}` executes `.toString()` successfully, while `Object.create(null)` throws a `TypeError: ...is not a function`.

D) Both objects throw a `ReferenceError` because `.toString()` is deprecated in modern JavaScript.

21) What is printed by the following program?

```
public class Main {  
    public static void main(String[] args) {  
        String str1 = "Java";  
        String str2 = new String("Java");  
        System.out.println((str1 == str2) + " " + str1.equals(str2));  
    }  
}
```

22) Predict the exact terminal output:

```
public class Test {  
    public static void main(String[] args) {  
        try {  
            System.out.print("X");  
            return;  
        } finally {  
            System.out.print("Y");  
        }  
    }  
}
```

23) What will be the output of this code snippet?

```
public class CacheTest {  
    public static void main(String[] args) {  
        Integer num1 = 100, num2 = 100;  
        Integer num3 = 300, num4 = 300;  
        System.out.println((num1 == num2) + " " + (num3 == num4));  
    }  
}
```

24) Analyze this array reference manipulation:

```
public class ArrayRef {  
    public static void main(String[] args) {  
        int[] original = {1, 2, 3};  
        int[] alias = original;
```

```
        alias[1] = 99;

        System.out.println(original[1]);
    }
}
```

25) What is the outcome of the following execution?

```
class Parent {

    static void display() { System.out.print("P"); }

}

class Child extends Parent {

    static void display() { System.out.print("C"); }

}

public class Test {

    public static void main(String[] args) {

        Parent obj = new Child();

        obj.display();

    }

}
```

26) What will be printed?

```
public class CharMath {

    public static void main(String[] args) {

        char ch = 'A';

        System.out.println(ch + 1);

    }

}
```

27) What is the output of this code?

```
public class DivisionTest {  
    public static void main(String[] args) {  
        System.out.println((10 / 4) + " " + (10 / 4.0));  
    }  
}
```

28) What will happen when compiling and executing this code?

```
public class FinalTest {  
    public static void main(String[] args) {  
        final int x;  
        x = 10;  
        System.out.println(x);  
    }  
}
```

29) What gets printed after modifying elements inside an enhanced for-loop?

```
public class LoopTest {  
    public static void main(String[] args) {  
        int[] vals = {10, 20, 30};  
        for (int v : vals) {  
            v = v * 2;  
        }  
        System.out.println(vals[1]);  
    }  
}
```


30) Predict the output of the static block execution:

```
public class StaticOrder {  
    static {  
        System.out.print("1");  
    }  
    public static void main(String[] args) {  
        System.out.print("2");  
    }  
}
```

31) What is the output of the following bitwise operation?

```
#include <stdio.h>  
  
int main() {  
    int a = 5, b = 2;  
    int res = a ^ b >> 1;  
    printf("%d\n", res);  
    return 0;  
}
```

32) Which bitwise expression evaluates to true ONLY if a non-zero integer N is a power of 2?

33) Predict the outcome of this pointer arithmetic snippet:

```
#include <stdio.h>  
  
int main() {  
    int arr[] = {10, 20, 30, 40, 50};  
    int *ptr = arr + 2;  
    printf("%d %d\n", *ptr, *(ptr - 1));  
}
```

```
    return 0;

}
```

34) What does sizeof(arr) evaluate to inside the function?

```
#include <stdio.h>

void checkSize(int arr[]) {

    printf("%zu\n", sizeof(arr));

}

int main() {

    int numbers[10];

    checkSize(numbers);

    return 0;

}
```

35) Assuming 4-byte integers and 1-byte chars on an x86-64 machine with standard alignment rules, what is sizeof(struct Demo)?

```
struct Demo {

    char c1;

    int num;

    char c2;

};
```

36) What is the output of the following code?

```
#include <stdio.h>

int* getValue() {

    static int val = 50;

    return &val;
```

```
}  
  
int main() {  
  
    int *p = getValue();  
  
    *p = 100;  
  
    printf("%d\n", *getValue());  
  
    return 0;  
  
}
```

37) What memory error occurs in the following code block?

```
#include <stdio.h>  
  
#include <stdlib.h>  
  
int main() {  
  
    int *ptr = (int*)malloc(sizeof(int));  
  
    *ptr = 42;  
  
    free(ptr);  
  
    printf("%d\n", *ptr);  
  
    return 0;  
  
}
```

38) What is printed by the following code?

```
#include <stdio.h>  
  
int main() {  
  
    int val = 8;  
  
    int count = 0;  
  
    while (val > 0) {  
  
        count += (val & 1);  
  
    }
```

```

        val >= 1;

    }

    printf("%d\n", count);

    return 0;

}

```

39) What is the standard function pointer syntax to declare a pointer named operation that takes two double arguments and returns an int?

Table 1 : Employees

emp_id	emp_name	dept_id	salary	manager_id
1	Alice	10	90000	NULL
2	Bob	10	70000	1
3	Charlie	20	80000	1
4	David	20	75000	3
5	Eve	30	50000	3

Table 2 : Departments

dept_id	dept_name	budget
10	Engineering	200000
20	Sales	150000
30	HR	60000
40	Marketing	100000

- 40) Write an SQL query to find the second-highest salary from the Employees table without using the LIMIT or TOP keywords.
- 41) Write an SQL query to retrieve the names of all employees along with their immediate manager's name. Include employees who do not have a manager.
- 42) Which aggregate query will return departments whose total employee salary payout exceeds 100,000?
- 43) Write a query to display dept_name for departments that currently have no employees assigned to them.
- 44) What will be the result of performing a FULL OUTER JOIN between Employees and Departments on dept_id in terms of total row count?
- 45) Write an SQL query using a subquery to list all employees who earn more than the average salary of their respective departments
- 46) What is the output of the following query?

```
SELECT COUNT(manager_id), COUNT(*)
```

```
FROM Employees;
```

- 47) Write a query to rank employees within each department based on salary in descending order using window functions.
- 48) What will be returned by this query?

```
SELECT e.emp_name, d.dept_name
```

```
FROM Employees e
```

```
LEFT JOIN Departments d ON e.dept_id = d.dept_id
```

```
WHERE d.budget > 100000;
```

- 49) Write a query to update the salaries of employees in the 'Engineering' department by giving them a 10% raise.
- 50) X and Y together can complete a project in 20 days. They work together for 12 days, after which Y leaves. X finishes the remaining work alone in 14 days. How many days would X take to complete the entire project working alone?
- 51) Train A leaves Station P at 6:00 AM and reaches Station Q at 10:00 AM. Train B leaves Station Q at 8:00 AM and reaches Station P at 11:30 AM. At what time do the two trains cross each other?
- 52) Identify the number that does not belong to the following sequence: 123, 234, 345, 456, 568, 678
- 53) P, Q, and R enter into a partnership with a total capital of ₹80,000. P invests ₹5,000 more than Q, and Q invests ₹10,000 more than R. Out of a total annual profit of ₹32,000, what is P's share?
- 54) A mother tells her daughter: "I was your current age when you were born." If the mother is currently 52 years old, how old was the daughter 6 years ago?

- 55) In how many distinct ways can the letters of the word "ORANGE" be arranged such that all vowels always remain together?
- 56) A 60-liter container contains a mixture of milk and water with 15% water content. How many liters of pure water must be added to make the water content 20% of the total mixture?
- 57) Referring to a boy in a photograph, a woman says: "He is the son of the only daughter of my mother's husband." How is the boy related to the woman?
- 58) A traveler starts at point O and walks 3 km West, then turns Right and walks 4 km, then turns Right and walks 6 km. How far and in which direction is the traveler relative to starting point O?
- 59) Find the missing term in the following numerical sequence: 4, 2, 2, 3, 6, 15, ?

